

Calculus & Optimization

- Functions of single variable
- Limit
- Continuity
- Differentiability
- Taylor Series
- Maxima & Minima
- Optimization involving single variable.

Syllabus

⇒ Functions of single variable:-

$$f(x) = 3x^2 + 2 \quad (\checkmark)$$

$$f(x, y) = 3x^2 + 2xy + 2y^2 + c \quad (\otimes)$$

⇒ Limit:- It tells us the value that a function approaches as the input gets closer to a certain number.

Ex:- If it gets close to $x=a$, what does $f(x)$ get close to?

→ Limit of $f(x)$ as x approaches " a ".

$$\lim_{x \rightarrow a} f(x)$$

Ex:- $f(x) = \lim_{x \rightarrow 2} (3x + 1)$

$$\left. \begin{aligned} f(1.9) &= (3 \times 1.9) + 1 = 6.7 \\ f(1.99) &= (3 \times 1.99) + 1 = 6.97 \end{aligned} \right\} \quad \begin{aligned} f(2) &= 3 \times 2 + 1 \\ &= \underline{\underline{7}} \end{aligned}$$

Ex:- $f(x) = \lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1}$

$$\left. \begin{aligned} f(0.9) &= 1.9 \\ f(0.99) &= 1.99 \\ f(1) &= \frac{1^2 - 1}{1 - 1} = \frac{0}{0} \end{aligned} \right\} \quad \begin{aligned} &\lim_{x \rightarrow a^-} f(x) \\ &\lim_{x \rightarrow a^+} f(x) \end{aligned} \quad \left[\begin{array}{l} \text{we can also} \\ \text{calculate using} \\ \lim_{x \rightarrow a} f(x) \end{array} \right]$$

(Indeterminate form)

7 Indeterminate forms

$\frac{0}{0}, \frac{\infty}{\infty}, 0 \times \infty, \infty - \infty, 1^\infty, 0^0, \infty^0$

L-Hospital Rule Convert then Log then

↑ ↑ ↑

L-Hospital's Rule :-

→ Evaluate limits that lead to Indeterminate forms.

$$\rightarrow \frac{0}{0}, \frac{\infty}{\infty}$$

$$\text{if } \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{0}{0} \text{ or } \frac{\infty}{\infty}$$

$$\text{then } \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

→ Replacing hard to evaluate functⁿ with their slopes, which often behave more nicely near trouble points.

$$\rightarrow \text{if } \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} = \frac{0}{0} \text{ or } \frac{\infty}{\infty}$$

then repeat.

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} = \lim_{x \rightarrow a} \frac{f''(x)}{g''(x)} = \dots$$

Ex:- $\lim_{x \rightarrow 0} \frac{\sin x}{x} = \frac{0}{0}$

$$\therefore \lim_{x \rightarrow 0} \frac{\sin x}{x} = \lim_{x \rightarrow 0} \frac{(\cos x)}{1} = \underline{\underline{1}}$$

$$\therefore \boxed{\lim_{x \rightarrow 0} \frac{\sin x}{x} = \lim_{x \rightarrow 0} \frac{\tan x}{x} = 1}$$

Ex:- 2 $\frac{\infty}{\infty}$ form

④

$$\lim_{x \rightarrow \infty} \frac{x}{e^x} = \frac{\infty}{\infty}$$

$$\therefore \lim_{x \rightarrow \infty} \frac{x}{e^x} = \lim_{x \rightarrow \infty} \frac{1}{e^x} = \frac{1}{\infty} = \underline{\underline{0}}$$

★ Ex:- 3

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{0}{0}$$

$$= \lim_{x \rightarrow 0} \frac{\sin x}{2x}$$

$$= \frac{1}{2} \because \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$\boxed{\lim_{x \rightarrow 0} \frac{\sin mx}{x} = \lim_{x \rightarrow 0} \frac{\tan mx}{x} = m}$$

① I^∞ form

$$\therefore \lim_{x \rightarrow 0} f(x)^{g(x)}$$

$$\lim_{x \rightarrow 0} f(x) \rightarrow 1$$

$$\lim_{x \rightarrow 0} g(x) \rightarrow \infty$$

$$y = \lim_{x \rightarrow 0} f(x)^{g(x)}$$

$$\ln y = \lim_{x \rightarrow 0} \frac{g(x) \ln[f(x)]}{\infty \cdot 0}$$

$$\frac{0 \times \infty}{1} \\ 0 \times \frac{1}{0} = \frac{0}{0}$$

$$\boxed{y = e^{\lim_{x \rightarrow a} g(x) [f(x) - 1]}}$$

② 0^0 form

$$\boxed{\lim_{x \rightarrow 0} f(x)^{f(x)} = 1}$$

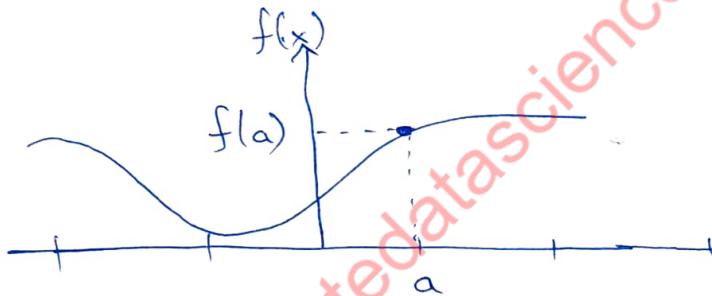
$$\text{if } f(x) = 0$$

⇒ Continuity :-

A function is continuous if it can be drawn without lifting a pen.

→ Continuity at a point :-

- 1) $f(a)$ is defined. (If $x \rightarrow a$ then $y \rightarrow$ some value)
- 2) $\lim_{x \rightarrow a} f(x)$ exists. $\left[\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) \right]$
- 3) $\lim_{x \rightarrow a} f(x) = f(a)$ $\left[\lim_{x \rightarrow a} f(x) = f(a) \right]$



→ Continuity on an Interval :-

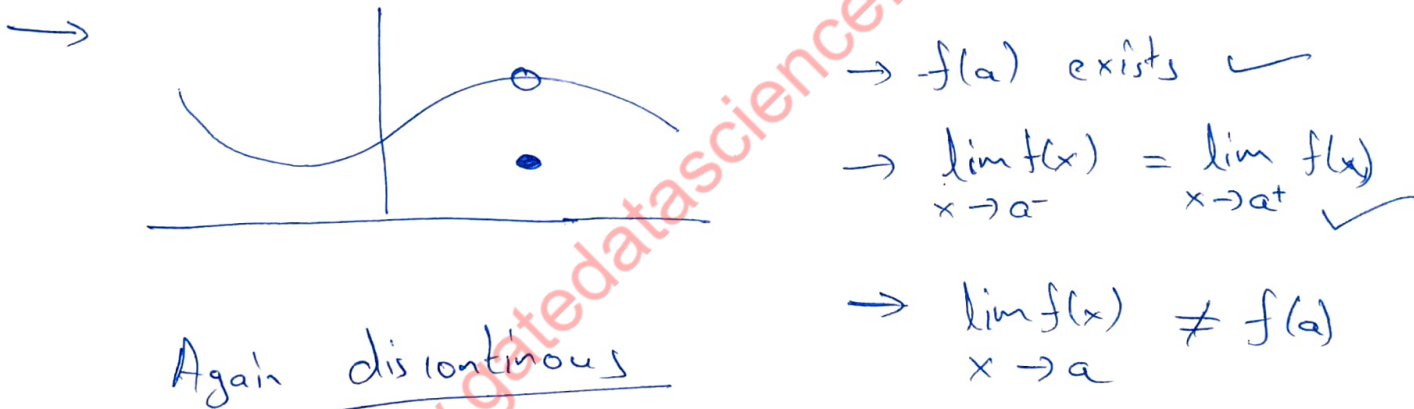
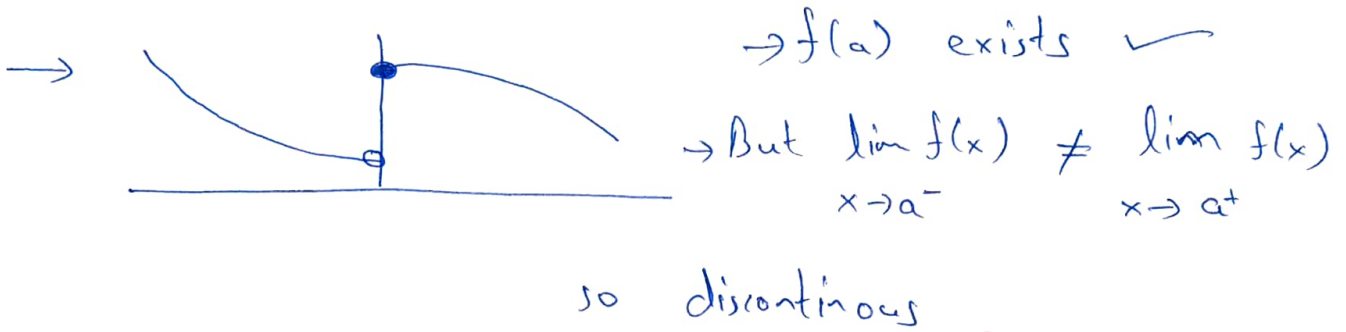
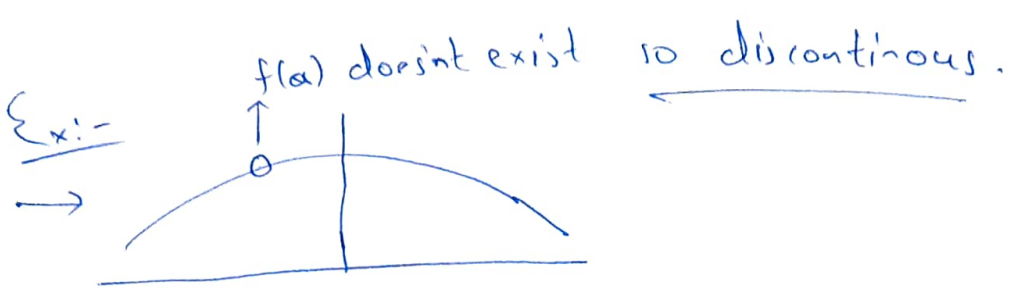
A function is continuous on an interval $[a, b]$ if it is continuous at every point in that interval.

- Open interval: (a, b)
- Closed interval: $[a, b]$ (check endpoints specially)

$$\hookrightarrow \lim_{x \rightarrow a^+} f(x) = f(a)$$

$$\lim_{x \rightarrow b^-} f(x) = f(b)$$

(6)



www.gatedata-science.com

⇒ Differentiability :-

(7)

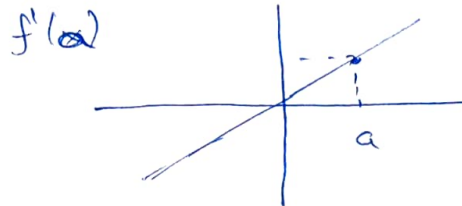
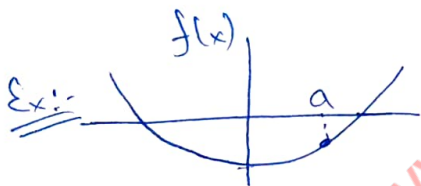
A function $f(x)$ is differentiable at a point $x = a$ if its derivative exists at that point.

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{(a+h) - (a)}$$

Note:- If $f(x)$ is differentiable, then it is automatically continuous, but reverse is not always true.

→ Differentiability at a Point:-

- $f'(a)$ should exist.



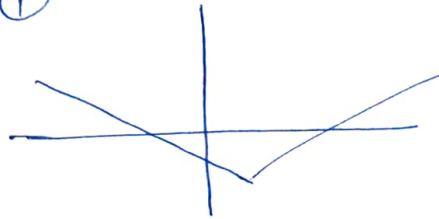
→ Differentiable at ' a '.

→ Differentiable on an Interval:-

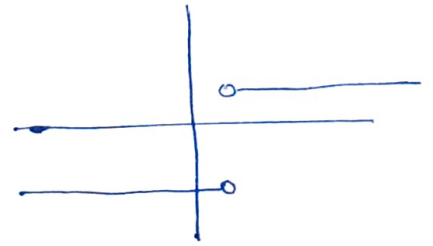
- Differentiable at every point on that interval
or its derivative should exist at every point.

Ex:- ①

$f(x)$



$f'(x)$

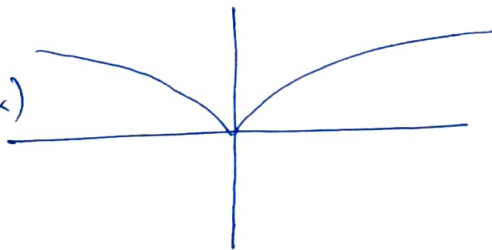


$f'(a)$ doesn't exist.

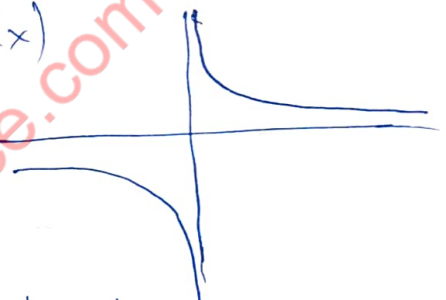
So Discontinuous :-

②

$f(x)$



$f'(x)$

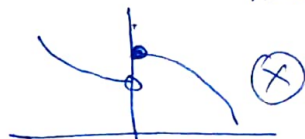


$f'(a)$ doesn't exist

So Discontinuous.

Recap :-

→ Continuous :- There should not be any gaps, Draw a functⁿ without picking up the Pen.



→ Differentiable :- functⁿ should be smooth without any sharp points or any instantaneous changes in the functⁿ slope.



⇒ Maxima & Minima :-

9

① → For funⁿ of One Variable $f(x)$

Method :-

1) Find $f'(x)$

2) Calculate Stationary pts (x_0) where $f'(x) = 0$

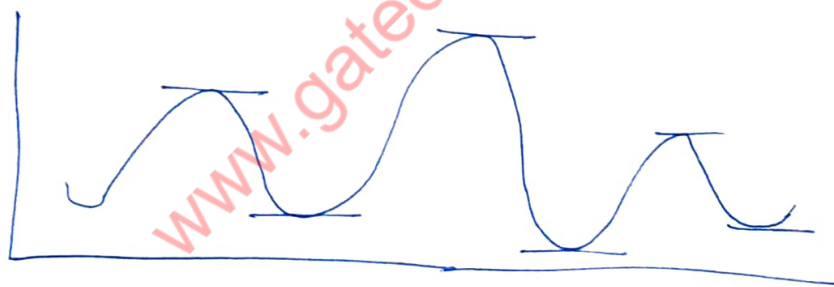
3) Find $f''(x)$

4) For each stationary pt find $f''(x)$

→ if $f''(x_0) > 0$ — local min

→ if $f''(x_0) < 0$ — local max

→ if $f''(x_0) = 0$ — neither max nor min.



→ To check Global min or Global Max in $[a, b]$

• Calculate $f[a], f[b], f[x_0]$

max will be global max

min will be global min

② → for functⁿ of 2 variables:-

$$z = f(x, y)$$

$$- p = \frac{\partial z}{\partial x}, \quad q = \frac{\partial z}{\partial y}$$

$$r = \frac{\partial^2 z}{\partial x^2}, \quad s = \frac{\partial^2 z}{\partial x \partial y}, \quad t = \frac{\partial^2 z}{\partial y^2}$$

- Put $p=0$ & $q=0$ for obtaining stationary pts.

- for each st. pts find r, s, t .

a) $rt - s^2 > 0$ & $r > 0 \rightarrow$ local min at stationary pt.

b) $rt - s^2 > 0$ & $r < 0 \rightarrow$ local max

c) $rt - s^2 < 0 \rightarrow$ No extreme
(SADDLE pt)

⇒ Taylor Series

→ Taylor series of $f(x)$ around $x=a$

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \dots$$

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n$$

if $a=0$ then
it's called
Maclaurin Series.

Expansion of e^x , $\sin x$, $\cos x$ at $a=0$

$$\rightarrow e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$\rightarrow \sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} + \dots$$

$$\rightarrow \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} + \dots$$

⇒ Optimization Involving Single Variable

- Process of finding the maximum or minimum values of a function
- Will solve as same as Maxima or Minima.